SHANKAR HONRAO

Pune, Maharashtra, India
+91 9373171998 | shankarhonrao70@gmail.com | linkedin.com/in/shankarhonrao |
github.com/shankarhonrao70

Professional Summary

Robotics and Automation Engineering student with hands-on experience in ROS 2, autonomous navigation, embedded systems, computer vision, and industrial automation. Skilled in developing intelligent robotic solutions using Python, C++, OpenCV, MediaPipe, Gazebo, and hardware integration. Passionate about autonomous systems, robotics software, and real-world engineering applications.

Technical Skills

Robotics & Simulation: ROS 2 Humble, SLAM, Autonomous Navigation, Gazebo, RViz
Programming: Python, C++, Arduino C/C++, MATLAB
Computer Vision: OpenCV, MediaPipe, Image Processing
Embedded Systems & Hardware: STM32, Arduino, IR Sensors, Ultrasonic Sensors, DC Motors, Servo Motors, Motor Drivers, Sensor Interfacing
CAD & Design: Fusion 360, SolidWorks, AutoCAD
Development Tools: Git, GitHub, Arduino IDE

Education

Bachelor of Technology (B.Tech) in Robotics & Automation Engineering
Zeal College of Engineering and Research, Pune
CGPA: 7.80/10

Projects

Gesture-Controlled TurtleBot3 (ROS 2 + Computer Vision)
- Developed a real-time hand gesture recognition system using OpenCV and MediaPipe.
- Integrated ROS 2 topics to convert hand gestures into robot velocity commands.
- Simulated and validated TurtleBot3 movement using Gazebo.

Autonomous Mobile Robot (ROS 2)
- Developed ROS 2 nodes for differential-drive robot navigation and control.
- Implemented and tested robot behavior using Gazebo and RViz simulation environments.

Line Following Robot (STM32 Based)
- Designed and developed an autonomous robot using IR sensors and STM32 microcontroller.
- Implemented sensor-based path detection and motor control for accurate navigation.

Smart Parking System (Arduino Based)
- Developed an automated parking slot detection system using Arduino and sensors.
- Implemented microcontroller logic for real-time parking availability monitoring.

Internship Experience

Robotic Software Intern – Simulation & Control Systems
NishCorp Technologies LLP — Jan 2026 – Feb 2026
- Worked on ROS 2 and Gazebo-based robotic simulation and testing.
- Developed motion control logic, robot kinematics, and basic autonomy modules.
- Assisted in debugging, performance optimization, and software-hardware integration.

Automation Intern
Saiprasad Enterprises (House of Automation), Pune
- Assisted in assembly of industrial X-Y linear motion systems for pick-and-place automation.
- Worked with linear rails, actuators, wiring, testing, and mechanical system integration.

Certifications & Achievements

- ROS 2 Completion Certificate – Hands-on training in ROS 2 architecture, nodes, topics, services, and robotic simulation.
- SolidWorks Workshop Certificate – Practical training in 3D CAD modeling, part design, and assembly.
- MATLAB Training Certificate – Engineering computation, programming, and data analysis using MATLAB.
- Arduino Competition Runner-Up – Secured runner-up position in an Arduino-based robotics and automation competition.
- e-Yantra Robotics Competition Participant – Participated in IIT Bombay's e-Yantra robotics competition involving robotics, embedded systems, and engineering problem-solving.